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Inductor component

Abstract

The inductor component includes a first magnetic layer, an inductor wiring line laminated on a main face of the first magnetic layer, a second magnetic layer disposed in the same layer as the inductor wiring line, a third magnetic layer disposed on a main face of the inductor wiring line and the second magnetic layer on a side opposite to the first magnetic layer, and an insulation layer being a non-magnetic material in contact with part of a surface of the inductor wiring line. The entire face of the surface of the inductor wiring line on the first magnetic layer side is in contact with the first magnetic layer, and a face of the surface of the inductor wiring line on the second magnetic layer side is in contact with the insulation layer.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims benefit of priority to Japanese Patent Application No. 2020-030656, filed Feb. 26, 2020, the entire content of which is incorporated herein by reference.

BACKGROUND

Technical Field

(2) The present disclosure relates to an inductor component.

Background Art

(3) An inductor wiring line is laminated on a surface of an insulation substrate in an inductor component described in Japanese Unexamined Patent Application Publication No. 2013-225718. The face of the inductor wiring line not covered by the insulation substrate is covered by an insulation layer. The outer face of the laminated structure constituted of the inductor wiring line, the insulation substrate, and the insulation layer is covered by a magnetic layer.

SUMMARY

(4) In the inductor component described in Japanese Unexamined Patent Application Publication No. 2013-225718, the proportion of the magnetic layer may be made larger as the proportion of the insulation substrate or the insulation layer is smaller when the overall volume of the inductor component is the same. This is advantageous in terms of improving the inductance. On the other hand, omitting all of the insulation substrate and the insulation layer may not ensure the demanded insulation property, and is not practical.

(5) Accordingly, one aspect of the present disclosure is an inductor component including a first magnetic layer; an inductor wiring line laminated on a main face of the first magnetic layer; a second magnetic layer disposed in a layer same as the inductor wiring line; a third magnetic layer disposed on a main face of the inductor wiring line and the second magnetic layer on a side opposite to the first magnetic layer; and an insulation layer being a non-magnetic material in contact with part of a surface of the inductor wiring line, in which an entire face of the surface of the inductor wiring line on a side of the first magnetic layer is in contact with the first magnetic layer, and a face of the surface of the inductor wiring line on the second magnetic layer side is in contact with the insulation layer.

(6) According to the configuration described above, the entire face of the surface of the inductor wiring line on the first magnetic layer side is in contact with the first magnetic layer without the insulation layer interposed therebetween. Absence of the insulation layer on the first magnetic layer side relative to the inductor wiring line allows the proportion of the insulation layer in the inductor component to be relatively small. Further, interposing the insulation layer in the same layer as the inductor wiring line makes it easy to ensure the insulation property in the same layer with the inductor wiring line. Note that the surface of the inductor wiring line refers to an outer face being a boundary with the outer side portion of the inductor wiring line.

(7) Other features, elements, characteristics and advantages of the present disclosure will become more apparent from the following detailed description of preferred embodiments of the present disclosure with reference to the attached drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an exploded perspective view of an inductor component in a first embodiment;
- (2) FIG. 2 is a top view of a second layer in the first embodiment;
- (3) FIG. 3 is a sectional view of the inductor component in the first embodiment taken along a line A-A in FIG. 2;
- (4) FIG. 4 is an enlarged sectional view of a contacting portion between an inductor wiring line and a magnetic layer in the first embodiment;
- (5) FIG. 5 is an explanatory diagram of a manufacturing method of the inductor component in the first embodiment;
- (6) FIG. 6 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (7) FIG. 7 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (8) FIG. 8 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (9) FIG. 9 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (10) FIG. 10 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (11) FIG. 11 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (12) FIG. 12 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (13) FIG. 13 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (14) FIG. 14 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (15) FIG. 15 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (16) FIG. 16 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (17) FIG. 17 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (18) FIG. 18 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (19) FIG. 19 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (20) FIG. 20 is an explanatory diagram of the manufacturing method of the inductor component in the first embodiment;
- (21) FIG. 21 is an exploded perspective view of an inductor component in a second embodiment;
- (22) FIG. 22 is a top view of a second layer in the second embodiment;
- (23) FIG. 23 is a sectional view of the inductor component in the second embodiment taken along a line B-B in FIG. 22;
- (24) FIG. 24 is an enlarged sectional view of the inductor component in the second embodiment;
- (25) FIG. 25 is an explanatory diagram of a manufacturing method of the inductor component in the second embodiment;
- (26) FIG. 26 is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;
- (27) FIG. 27 is an explanatory diagram of the manufacturing method of the inductor component in

the second embodiment;

(28) FIG. **28** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(29) FIG. **29** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(30) FIG. **30** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(31) FIG. **31** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(32) FIG. **32** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(33) FIG. **33** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(34) FIG. **34** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(35) FIG. **35** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(36) FIG. **36** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(37) FIG. **37** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(38) FIG. **38** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(39) FIG. **39** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(40) FIG. **40** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment;

(41) FIG. **41** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment; and

(42) FIG. **42** is an explanatory diagram of the manufacturing method of the inductor component in the second embodiment.

DETAILED DESCRIPTION

(43) Hereinafter, an inductor component and an embodiment of the inductor component will be described. Note that constituent elements may be illustrated in an enlarged manner in order to facilitate understanding of the drawings. The dimensional ratio of the constituent element may differ from the actual dimensional ratio or that in another drawing.

First Embodiment

(44) Hereinafter, a first embodiment of the inductor component will be described.

(45) An inductor component **10** has, as a whole, a structure in which three thin plate-shape layers are laminated in a thickness direction as illustrated in FIG. **1**. In the following description, a lamination direction of each of three layers will be described as an up-down direction.

(46) A first layer **L1** has a substantially square shape when viewed in the up-down direction. The first layer **L1** is constituted of only a first magnetic layer **21**. The first magnetic layer **21** is a mixture of a resin and a magnetic metal powder, and is a magnetic body as a whole. Magnetic metal powders **20B** are dispersed in a base material **20A** made of an insulation material in the first magnetic layer **21** as illustrated in FIG. **4**. The first magnetic layer **21**, therefore, is a magnetic material as a whole. The base material **20A** is composed of an epoxy-based resin and an inorganic filler having an average particle size equal to or less than about 1.0 μm . Further, the magnetic metal powder **20B** is an alloy made of iron, silicon, and chromium, and the average particle size of the magnetic metal powder **20B** is equal to or less than about 5.0 μm . In the present embodiment, the

first layer **L1** is the lowermost layer in the up-down direction. That is, in the up-down direction, the side on which an outer electrode **70** is provided is referred to as an upper side, and the opposite side thereof is referred to as a lower side. The outer electrode **70** will be described later.

(47) A second layer **L2** having the substantially square shape same as the first layer **L1** when viewed in the up-down direction is laminated on the upper side face of the first layer **L1** in the lamination direction as illustrated in FIG. **1**. In the embodiment, the face of the second layer **L2** contacting with the first layer **L1** is a main face **MF** of the second layer **L2**. The second layer **L2** is constituted of an inductor wiring line **30**, a first dummy wiring line **41**, a second dummy wiring line **42**, a second magnetic layer **22**, and a first insulation portion **81**. That is, the inductor wiring line **30** constituting part of the second layer **L2** is laminated on a main face of the first magnetic layer **21** constituting the first layer **L1**.

(48) The inductor wiring line **30** is constituted of a wiring line main body **31**, a first pad **32**, and a second pad **33** in the second layer **L2** as illustrated in FIG. **2**. The inductor wiring line **30** extends in a spiral shape around the center of a substantially square shape in the second layer **L2** when viewed from the upper side in the up-down direction. Specifically, when viewed from the upper side in the up-down direction, the wiring line main body **31** of the inductor wiring line **30** is spirally wound in a counterclockwise direction from an outer peripheral end portion **31A** in the outer side portion in a radial direction toward an inner peripheral end portion **31B** in the inner side portion in the radial direction.

(49) The number of turns of the inductor wiring line **30** is determined based on a virtual vector. The starting point of the virtual vector is placed on a virtual center line extending along the extending direction of the inductor wiring line **30** through the center of the wiring line width of the inductor wiring line **30**. The direction of the virtual vector rotates in a normal direction view when the starting point of the inductor wiring line **30** is moved from one end to the other end of the virtual center line. The number of turns of the inductor wiring line **30** is defined as about 1.0 turns when the direction of the virtual vector rotates about 360 degrees. Thus, when the inductor wiring line **30** is wound about 180 degrees, the number of turns is about 0.5 turns, for example. The direction of the virtual vector virtually placed on the inductor wiring line rotates about 540 degrees in the embodiment. With this, the number of turns of the wound inductor wiring line **30** is about 1.5 turns in the embodiment.

(50) The first pad **32** is connected to the outer peripheral end portion **31A** of the wiring line main body **31**. The first pad **32** has a substantially circular shape when viewed in the up-down direction. The diameter of the circle of the first pad **32** is larger than the wiring line width of the wiring line main body **31**.

(51) The first dummy wiring line **41** extends from the first pad **32** toward the outer edge side of the second layer **L2**. The first dummy wiring line **41** extends to the side face of the second layer **L2**, and is exposed to the outer face of the inductor component **10**.

(52) The second pad **33** is connected to the inner peripheral end portion **31B** of the wiring line main body **31**. The second pad **33** has a substantially circular shape when viewed in the up-down direction. The diameter of the circle of the second pad **33** is larger than the wiring line width of the wiring line main body **31**.

(53) The second dummy wiring line **42** extends from a portion wound by 0.5 turns from the outer peripheral end portion **31A** in a region between the outer peripheral end portion **31A** and the inner peripheral end portion **31B** of the wiring line main body **31**. The second dummy wiring line **42** extends to the side face of the second layer **L2**, and is exposed to the outer face of the inductor component **10**. The inductor wiring line **30**, the first dummy wiring line **41**, and the second dummy wiring line **42** are integrated with one another in the embodiment.

(54) The inductor wiring line **30** has a laminated structure including a catalyst layer **30A**, a first wiring line layer **30B**, and a second wiring line layer **30C** in order from the side of the first magnetic layer **21** constituting the first layer **L1** as illustrated in FIG. **4**. The catalyst layer **30A** of

the inductor wiring line **30** is in contact with the upper face of the first magnetic layer **21**, and constitutes a main face MF of the second layer L2. The material of the catalyst layer **30A** is palladium. Only the inductor wiring line **30** and the first magnetic layer **21** that is described above are illustrated in FIG. 4, and other constituent elements are not illustrated.

(55) The first wiring line layer **30B** is directly laminated on the upper face of the catalyst layer **30A**. The material of the first wiring line layer **30B** has a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt %. A thickness TB of the first wiring line layer **30B** is equal to or less than about one-tenth of the wiring line width of the inductor wiring line **30**. The thickness TB of the first wiring line layer **30B** is about 2.0 μm in the embodiment. The thickness TB of the first wiring line layer **30B** is determined as follows. The thickness from the upper end of the first magnetic layer **21** to the upper end of the first wiring line layer **30B** is measured at three points in a cross section along the lamination direction in one observation field under a microscope of 1500 times magnification. The thickness TB of the first wiring line layer **30B** is determined as the average value of the three measured values. The thickness TB of the first wiring line layer **30B** is substantially constant in the embodiment. Note that the thickness of the catalyst layer **30A** described above is exaggerated in FIG. 4, but is much smaller than the thickness of the first wiring line layer **30B**. Thus, in measuring the thickness TB of the first wiring line layer **30B**, measuring the thickness from the upper end of the first magnetic layer **21**, that is, including the thickness of the catalyst layer **30A**, does not affect the measurement. However, when the interface of the catalyst layer **30A** is clearly confirmed, the thickness TB may be measured from the upper face of the catalyst layer **30A**. The wiring line width of the inductor wiring line **30** is determined as the average value of three points of the width dimension of the inductor wiring line **30** in the vicinity of the center in the extending direction.

(56) The second wiring line layer **30C** is directly laminated on the upper face of the first wiring line layer **30B** being the surface on a third magnetic layer **23** side. A thickness TC of the second wiring line layer **30C** is equal to or about five times larger than the thickness TB of the first wiring line layer **30B**. The second wiring line layer **30C** has the thickness TC of about 45 μm in the embodiment. An overall thickness TA of the inductor wiring line **30** is a total value of the thickness TB of the first wiring line layer **30B** and the thickness TC of the second wiring line layer **30C** as illustrated in FIG. 3, and is about 47 μm. The material of the second wiring line layer **30C** has a copper ratio equal to or larger than about 99 wt %, and the nickel ratio is equal to or less than the detection limit.

(57) An anchor portion **34** extends from the main face MF of the inductor wiring line **30** as illustrated in FIG. 4. The anchor portion **34** covers the surfaces of the magnetic metal powders **20B** in contact with the main face MF among the large number of magnetic metal powders **20B** in the first magnetic layer **21**. The anchor portion **34** extends from the main face MF so as to enter a gap between the base material **20A** and the magnetic metal powder **20B** in the first magnetic layer **21**. Further, the magnetic metal powder **20B** covered by the anchor portion **34** includes a cross section in which equal to or more than about one-third of the surface is covered by the anchor portion **34** when the magnetic metal powder **20B** is viewed in a cross section. The cross section is orthogonal to the main face MF in the embodiment.

(58) In the inductor wiring line **30** of the second layer L2, a 0.5 turns portion on the first pad **32** side and a 0.5 turns portion on the second pad **33** side extend in parallel to each other as illustrated in FIG. 1. The distance between the wiring lines in the inductor wiring line **30** is minimum between the inner side face of the 0.5 turns portion in the radial direction on the first pad **32** side and the outer side face of the 0.5 turns portion in the radial direction on the second pad **33** side. Further, in the second layer L2, a first insulation portion **81** is interposed between the inner side face of the 0.5 turns portion in the radial direction on the first pad **32** side and the outer side face of the 0.5 turns portion in the radial direction on the second pad **33** side. That is, the first insulation portion **81** is interposed in the portion where the distance between the wiring lines in the inductor wiring line **30**

is minimum, and extends in a substantially arc shape along the inductor wiring line **30**. The first insulation portion **81** is composed of an epoxy-based resin and an inorganic filler having an average particle size equal to or less than about 1.0 μm .

(59) In the second layer L2, the portion other than the inductor wiring line **30**, the first dummy wiring line **41**, the second dummy wiring line **42**, and the first insulation portion **81** is the second magnetic layer **22**. Accordingly, the second magnetic layer **22** is present in the central portion of the second layer L2 and in the portion of the second layer L2 on the outer side relative to the inductor wiring line **30**. Therefore, a face of the surface of the inductor wiring line **30** on the side opposite to the first insulation portion **81** is in contact with the second magnetic layer **22**. In addition, the material of the second magnetic layer **22** is the same as that of the first magnetic layer **21**. As described above, the second magnetic layer **22** is disposed in the same layer as the inductor wiring line **30**.

(60) A third layer L3 having the substantially square shape same as the second layer L2 when viewed in the up-down direction is laminated on the upper face of the second layer L2. The third layer L3 is constituted of a first vertical wiring line **51**, a second vertical wiring line **52**, the third magnetic layer **23**, and a second insulation portion **82**. The first vertical wiring line **51** and the second vertical wiring line **52** are connected to the inductor wiring line **30**, and penetrate through the third magnetic layer **23** from the main face on the inductor wiring line **30** side toward the main face on the side opposite to the inductor wiring line **30** side.

(61) The first vertical wiring line **51** is directly connected to the upper side face of the first pad **32** without any other layers interposed therebetween. The first vertical wiring line **51** has a substantially columnar shape, and an axis line direction of the column coincides with the up-down direction. The diameter of the substantially circular first vertical wiring line **51** is smaller than the diameter of the first pad **32** when viewed in the up-down direction. The material of the first vertical wiring line **51** is the same as that of the second wiring line layer **30C** of the inductor wiring line **30**.

(62) The second vertical wiring line **52** is directly connected to the upper side face of the second pad **33** without any other layers interposed therebetween. The second vertical wiring line **52** has a substantially columnar shape, and an axis line direction of the column coincides with the up-down direction. The diameter of the substantially circular second vertical wiring line **52** is smaller than the diameter of the second pad **33** when viewed in the up-down direction. The material of the second vertical wiring line **52** is the same as that of the second wiring line layer **30C** of the inductor wiring line **30**. Note that the second wiring line layer **30C** of the inductor wiring line **30**, the first dummy wiring line **41**, the second dummy wiring line **42**, the first vertical wiring line **51**, and the second vertical wiring line **52** are integrated with one another. The first vertical wiring line **51** and the second vertical wiring line **52** are virtually illustrated by a dashed-and-double-dotted line in FIG. 2.

(63) The second insulation portion **82** is directly connected to the upper side face of the first insulation portion **81** without any other layers interposed therebetween as illustrated in FIG. 1. The second insulation portion **82** covers a wider range than the first insulation portion **81** in the width direction orthogonal to the extending direction of the first insulation portion **81**. As a result, the second insulation portion **82** covers part of the upper face of the 0.5 turns portion on the first pad **32** side and part of the upper face of the 0.5 turns portion on the second pad **33** side in the inductor wiring line **30**. The thickness of the second insulation portion **82** is smaller than the thickness of the third layer L3, and the third magnetic layer **23** is laminated on the upper side of the second insulation portion **82** in the third layer L3. The second insulation portion **82** is composed of epoxy-based resin and an inorganic filler having an average particle size equal to or less than about 1.0 μm , similarly to the first insulation portion **81**. Note that an insulation layer **80** is constituted of the first insulation portion **81** and the second insulation portion **82** in the embodiment. The second insulation portion **82** is virtually illustrated by a dashed-and-double-dotted line in FIG. 2.

(64) In the third layer L3, the portion other than the second insulation portion **82**, the first vertical

wiring line **51**, and the second vertical wiring line **52** is the third magnetic layer **23** as illustrated in FIG. **1**. The third magnetic layer **23**, therefore, is laminated on an upper side main face of the inductor wiring line **30** and the second magnetic layer **22** in the lamination direction on the side opposite to the first magnetic layer **21**.

(65) A covering layer **60** made of an insulation material is laminated on an upper side main face of the surface of the third layer **L3** on the side opposite to the inductor wiring line **30** as illustrated in FIG. **3**. The covering layer **60** covers substantially the entire area of the upper side face of the third layer **L3**, but holes are formed in portions corresponding to the first vertical wiring line **51** and the second vertical wiring line **52** in the third layer **L3**. That is, the upper side face of the first vertical wiring line **51** and the second vertical wiring line **52** on the side opposite to the inductor wiring line **30** is not covered by the covering layer **60**.

(66) An outer electrode **70** is connected to an upper side face of the first vertical wiring line **51**. The outer electrode **70** seems as if it is penetrating through the covering layer **60**, and the upper face of the outer electrode **70** is exposed from the covering layer **60**. The outer electrode **70** has a three-layer structure, and is constituted of a copper layer **70A**, a nickel layer **70B**, and a gold layer **70C** in order from the lower side in the lamination direction. In addition, the outer electrode **70** is also connected to the upper side face of the second vertical wiring line **52**. Note that the covering layer **60** and the outer electrode **70** are not illustrated in FIG. **1**.

(67) Next, a manufacturing method of the inductor component **10** in the first embodiment will be described.

(68) The manufacturing method of the inductor component **10** includes a first magnetic layer processing step, a first covering step, an inductor wiring line processing step, a first resist layer removing step, an insulation layer processing step, a second covering step, a vertical wiring line processing step, a second resist layer removing step, a second magnetic layer processing step, a covering layer processing step, a base substrate removing step, an outer electrode processing step and a dividing step as illustrated in FIG. **5**.

(69) In manufacturing the inductor component **10**, first, the first magnetic layer processing step is performed. A base substrate with a copper foil **95** is prepared as illustrated in FIG. **6**. A base substrate **96** of the base substrate with the copper foil **95** has a plate-shape. A copper foil **97** is laminated on the upper side face of the base substrate **96** in the lamination direction. Then, the first magnetic layer **21** composed of the base material **20A** and the magnetic metal powder **20B** is formed on the upper side face of the copper foil **97** of the base substrate with copper foil **95** as illustrated in FIG. **7**. In forming the first magnetic layer **21**, an insulation resin containing the magnetic metal powder **20B** is applied, and the insulation resin is solidified by press working to obtain the base material **20A**. The upper portions of the base material **20A** and the magnetic metal powder **20B** are ground such that the thickness in the up-down direction of the first magnetic layer **21** becomes a desired thickness. It is preferable to form a slight gap at the interface between the base material **20A** and the magnetic metal powder **20B** by controlling process parameters during the grinding. For example, vibrating the magnetic metal powder **20B** exposed from the base material **20A** by the grinding tool may form a slight gap between the base material **20A** and the magnetic metal powder **20B**. More specifically, the grinding tool is in contact with the base material **20A** and the magnetic metal powder **20B** to grind the upper portions of the base material **20A** and the magnetic metal powder **20B**, and an appropriate vibration of the grinding tool causes the vibration of the magnetic metal powder **20B** to be larger because the magnetic metal powder **20B** is harder than the base material **20A** made of an insulation resin. As described above, a slight gap is formed by the difference in vibration between the base material **20A** and the magnetic metal powder **20B**.

(70) The first covering step is performed next to the first magnetic layer processing step. In the first covering step, a first resist layer **91** covering the portion of the upper side face of the first magnetic layer **21**, on which the inductor wiring line **30**, the first dummy wiring line **41**, and the second

dummy wiring line **42** are not formed, is patterned by photolithography as illustrated in FIG. **8**. Specifically, first, a photosensitive dry film resist is applied to the entire upper side face of the first magnetic layer **21**. Next, the portion of the upper side face of the first magnetic layer **21**, on which the inductor wiring line **30**, the first dummy wiring line **41**, and the second dummy wiring line **42** are not formed, is exposed to light. As a result, of the applied dry film resist, the portion exposed to light is cured. Thereafter, the uncured portion of the applied dry film resist is peeled and removed by a chemical solution. Thus, the cured portion of the applied dry film resist is formed as the first resist layer **91**. On the other hand, the first magnetic layer **21** is exposed in the portion where the applied dry film resist is removed by a chemical solution and is not covered by the first resist layer **91**.

(71) An inductor wiring line processing step is performed next to the first covering step. In the inductor wiring line processing step, the inductor wiring line **30** constituted of the catalyst layer **30A**, the first wiring line layer **30B**, and the second wiring line layer **30C** is formed on the upper side face of the first magnetic layer **21**. Specifically, first, a portion which is not covered by the first resist layer **91** is made to adsorb palladium as illustrated in FIG. **9** in the upper side face of the first magnetic layer **21**. With this, the palladium adsorbed on the upper side face of the first magnetic layer **21** is formed as the catalyst layer **30A**. Next, the electroless copper plating by performing the immersion in an electroless copper plating solution forms the first wiring line layer **30B** having a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt % on the upper side face of the catalyst layer **30A**. The electroless copper plating solution is an alkaline solution, and contains copper salts such as copper chloride, copper sulfate and the like. On the other hand, the material of the magnetic metal powder **20B** is iron, and the ionization tendency is larger than that of copper being the material of the first wiring line layer **30B**. Therefore, in the inductor wiring line processing step, iron on the surface of the magnetic metal powder **20B** dissolves, and instead, copper is deposited on the surface of the magnetic metal powder **20B**.

(72) Here, since the electroless copper plating solution enters a slight gap between the base material **20A** and the magnetic metal powder **20B**, the substitution of iron with copper occurs not only on the exposed face side of the magnetic metal powder **20B** but also on the surface of the magnetic metal powder **20B** on the inner side of the base material **20A**. Copper deposited on the surface of the magnetic metal powder **20B** on the inner side of the base material **20A** functions as the anchor portion **34**.

(73) The covering amount is controlled such that copper deposited on the surface of the magnetic metal powder **20B** on the inner side of the base material **20A** covers equal to or more than about one-third of the surface area of the magnetic metal powder **20B**. Specifically, the covering amount is controlled by such as the application duration of a voltage for the electroless copper plating, the amount of the electric current, the content of copper or the catalyst in the plating solution or the like.

(74) An electrolytic copper plating is performed next to the electroless copper plating as illustrated in FIG. **10**. With this, the second wiring line layer **30C** having a copper ratio equal to or larger than about 99 wt % is formed on the surface of the first wiring line layer **30B**. As described above, the inductor wiring line **30** is formed by the adsorption of palladium, the electroless copper plating, and the electrolytic copper plating.

(75) The first resist layer removing step for removing the first resist layer **91** is performed next to the inductor wiring line processing step. In the first resist layer removing step, the first resist layer **91** is peeled off to remove from the first magnetic layer **21** as illustrated in FIG. **11**.

(76) The insulation layer processing step is performed next to the first resist layer removing step. In the insulation layer processing step, first, the insulation resin is applied to the upper side faces of the first magnetic layer **21**, the inductor wiring line **30**, the first dummy wiring line **41**, and the second dummy wiring line **42** as illustrated in FIG. **12**. Next, a portion for forming the first insulation portion **81** and the second insulation portion **82** is exposed to light as illustrated in FIG.

13. As a result, the portion exposed to light is cured. Thereafter, the uncured portion of the insulation resin is peeled and removed by a chemical solution. Thus, the cured portion of the applied insulation resin is formed as the insulation layer **80**.

(77) The second covering step is performed next to the insulation layer processing step. In the second covering step, patterned is a second resist layer **92** covering a portion of the upper side face of the first magnetic layer **21** and the upper side face of the second wiring line layer **30C**, on which the first vertical wiring line **51** and the second vertical wiring line **52** are not formed as illustrated in FIG. **14**. Note that the aspect of the photolithography in the second covering step is the same as that in the first covering step and a detailed description thereof will be omitted.

(78) The vertical wiring line processing step for forming the first vertical wiring line **51** and the second vertical wiring line **52** is performed next to the second covering step. In the vertical wiring line processing step, electrolytic copper plating is performed, and the first vertical wiring line **51** and the second vertical wiring line **52** are formed in a portion which is not covered by the second resist layer **92** of the upper side face of the second wiring line layer **30C**. The first vertical wiring line **51** and the second vertical wiring line **52** have the copper ratio equal to or larger than about 99 wt %.

(79) The second resist layer removing step for removing the second resist layer **92** is performed next to the vertical wiring line processing step as illustrated in FIG. **15**. In the second resist layer removing step, the second resist layer **92** is peeled off to remove from the first magnetic layer **21** similarly to the first resist layer removing step.

(80) The second magnetic layer processing step is performed next to the second resist layer removing step. In the second magnetic layer processing step, first, a magnetic material is filled from the upper side face of the first magnetic layer **21** toward the upper side relative to the upper ends of the first vertical wiring line **51** and the second vertical wiring line **52** in the lamination direction as illustrated in FIG. **16**. Next, the second magnetic layer **22** and the third magnetic layer **23** are formed by grinding the magnetic material from the upper side in the lamination direction until the upper ends of the first vertical wiring line **51** and the second vertical wiring line **52** are exposed.

(81) The covering layer processing step is performed next to the second magnetic layer processing step. In the covering layer processing step, a solder resist functioning as the covering layer **60** is patterned by photolithography on the portion on which the outer electrode **70** is not formed among the portions of the upper side face of the third magnetic layer **23**, the upper side face of the first vertical wiring line **51**, and the upper side face of the second vertical wiring line **52**, as illustrated in FIG. **17**.

(82) The base substrate removing step is performed next to the covering layer processing step. In the base substrate removing step, the base substrate with the copper foil **95** is removed as illustrated in FIG. **18**. Specifically, the base substrate **96** is peeled off to remove from the first magnetic layer **21**. Next, the copper foil **97** is removed by etching. Then, the first magnetic layer **21** is ground from the lower side in the lamination direction until the thickness from the lower end of the first magnetic layer **21** to the upper end of the covering layer **60** reaches a desired value.

(83) The outer electrode processing step is performed next to the base substrate removing step. The outer electrode **70** is formed on the upper side face of the first vertical wiring line **51** as illustrated in FIG. **19**. Further, the outer electrode **70** is formed on the upper side face of the second vertical wiring line **52**. In the outer electrode **70**, the copper layer **70A**, the nickel layer **70B**, and the gold layer **70C** are formed by electroless plating for copper, nickel, and gold, respectively. Thus, the outer electrode **70** having a three-layer structure is formed.

(84) The dividing step is performed next to the outer electrode processing step. Specifically, the dividing is performed by cutting with a dicing machine at break lines DL as illustrated in FIG. **20**. Thus, the inductor component **10** may be obtained. In addition, at this time, the first dummy wiring line **41** and the second dummy wiring line **42** included in the break lines DL are exposed to the side

faces of the inductor component **10**.

(85) Next, effects of the first embodiment will be described.

(86) (1) According to the inductor component **10** in the first embodiment, in the second layer L2 in which the inductor wiring line **30** is disposed, the insulation layer **80** is disposed on part of the side face of the inductor wiring line **30**. Therefore, the insulation property between the wiring lines in the inductor wiring line **30** may be ensured. On the other hand, the face of the surface of the inductor wiring line **30** on the first magnetic layer **21** side is not covered by the insulation layer **80**, and is in contact with the first magnetic layer **21**. The proportion of the insulation layer **80** in the inductor component **10** may be reduced and the proportion of the first magnetic layer **21** in the inductor component **10** may be increased because the insulation layer **80** does not exist on the face on the first magnetic layer **21** side. Therefore, when the volume of the inductor component **10** is the same, it is advantageous in terms of inductance because the proportion of the first magnetic layer **21** is large.

(87) (2) According to the inductor component **10** in the first embodiment, the number of turns of the inductor wiring line **30** is about 1.5 turns. Since the number of turns of the inductor wiring line **30** is more than about 1.0 turns, there is an area in which the wiring lines in the inductor wiring line **30** become close to each other in some extent. According to the first embodiment described above, the insulation layer **80** is interposed between the wiring lines in the inductor wiring line **30** where the distance between the wiring lines in the inductor wiring line **30** is the smallest. Therefore, it is possible to ensure the insulation property at a portion where the wiring lines in the inductor wiring line **30** are likely to be short-circuited.

(88) (3) According to the inductor component **10** in the first embodiment, part of the face of the inductor wiring line **30** on the third magnetic layer **23** side is covered by the insulation layer **80**. Therefore, the insulation property with the side of the third magnetic layer **23** on which the outer electrode **70** is disposed may easily be ensured.

(89) (4) According to the inductor component **10** in the first embodiment, the thickness TC of the second wiring line layer **30C** is equal to or larger than about five times the thickness TB of the first wiring line layer **30B**. Therefore, the thickness TA of the inductor wiring line **30** may be increased to some extent and a DC resistance may be reduced.

(90) (5) According to the manufacturing method of the inductor component **10** in the first embodiment, the electrolytic copper plating is performed, and the second wiring line layer **30C** is formed on the surface of the first wiring line layer **30B**. The second wiring line layer **30C** has the copper ratio equal to or larger than about 99 wt % and the nickel ratio equal to or less than the detection limit. Therefore, the second wiring line layer **30C** having a large thickness may be efficiently formed compared with the electroless copper plating.

(91) (6) According to the inductor component **10** in the first embodiment, the catalyst layer **30A** is disposed on the first magnetic layer **21** side of the first wiring line layer **30B**. The catalyst layer **30A** activates the deposition of copper in the electroless copper plating. Therefore, since palladium as the catalyst is adsorbed in a layered manner on the entire surface of the first magnetic layer **21**, copper is deposited on the entire surface of the first magnetic layer **21** when the electroless copper plating is performed. This makes it easy to form the first wiring line layer **30B** having a uniform thickness.

(92) (7) According to the inductor component **10** in the first embodiment, the covering layer **60** covers the upper face of the third layer L3. Therefore, the insulation property with the external factors may easily be ensured.

(93) (8) According to the inductor component **10** in the first embodiment, the insulation layer **80** contains epoxy-based resin and an inorganic filler. Therefore, a physical defect such as a crack is unlikely to occur when the thickness of the first magnetic layer **21** is reduced to some extent, and sufficient strength may be maintained without an additive insulation substrate or the like.

(94) (9) According to the inductor component **10** in the first embodiment, the anchor portion **34**

extends from the lower side face of the catalyst layer **30A** constituting the main face MF of the inductor wiring line **30**. The anchor portion **34** covers the surfaces of the magnetic metal powders **20B** dispersed in the base material **20A** of the first magnetic layer **21**. Therefore, an anchor effect is obtained between the inductor wiring line **30** and the first magnetic layer **21** by the anchor portion **34**. As a result, adhesion between the inductor wiring line **30** and the first magnetic layer **21** is improved. Thus, in the inductor component **10**, it is achieved that the inductor wiring line **30** is directly laminated on the first magnetic layer **21** while ensuring the required adhesion between the inductor wiring line **30** and the first magnetic layer **21**.

(95) (10) According to the inductor component **10** in the first embodiment, the magnetic metal powder **20B** covered by the anchor portion **34** includes a cross section in which equal to or more than about one-third of the surface is covered by the anchor portion **34** when the magnetic metal powder **20B** is viewed in a cross section. Therefore, the relatively large anchor portion **34** allows reliable close contact between the inductor wiring line **30** and the first magnetic layer **21**.

(96) (11) According to the manufacturing method of the inductor component **10** in the first embodiment, the magnetic metal powder **20B** is exposed on part of the surface of the first magnetic layer **21** in the inductor wiring line processing step, and the inductor wiring line **30** is formed on part of the surface of the first magnetic layer **21** by immersing the first magnetic layer **21** in a plating solution. Therefore, the plating liquid enters the gap between the base material **20A** and the magnetic metal powder **20B** in the first magnetic layer **21**, and the anchor portion **34** may be formed on the surface of the magnetic metal powder **20B** on the inner side of the base material **20A**.

(97) (12) According to the manufacturing method of the inductor component **10** in the first embodiment, the electroless copper plating is performed by performing immersion in the electroless copper plating solution, and the first wiring line layer **30B** having a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt % is formed on the upper side face of the catalyst layer **30A**. Therefore, damage to the surface of the first magnetic layer **21**, in a case where the first wiring line layer **30B** is formed by such as sputtering for example, is made relatively small, and the first wiring line layer **30B** may be formed without excessively reducing the amount of the magnetic metal powder **20B** in the first magnetic layer **21**.

(98) (13) According to the inductor component **10** in the first embodiment, iron being a material of the magnetic metal powder **20B** has a higher ionization tendency than copper being the material of the first wiring line layer **30B**. Therefore, iron having a large ionization tendency becomes ions and copper having a small ionization tendency deposits between the copper salt in the electroless copper plating and the surface of the magnetic metal powder **20B**. With this, it is possible for copper to deposit so as to cover the surface of the magnetic metal powder **20B** even when the base material **20A** and the magnetic metal powder **20B** are relatively dense.

Second Embodiment

(99) Hereinafter, a second embodiment of the inductor component will be described.

(100) An inductor component **110** has, as a whole, a structure in which six plate-shape layers are laminated in a thickness direction as illustrated in FIG. **21**. In the following description, the lamination direction in which six layers each are laminated will be described as an up-down direction.

(101) A first layer **L11** has a substantially rectangular shape when viewed in the up-down direction. The first layer **L11** is constituted of only a first magnetic layer **121**. Magnetic metal powders **120B** are dispersed in a base material **120A** made of an insulation material in the first magnetic layer **121** as illustrated in FIG. **24**. The first magnetic layer **121**, therefore, is a magnetic material as a whole. Specifically, the base material **120A** is composed of epoxy-based resin and an inorganic filler having an average particle size equal to or less than about 1.0 μm , and the magnetic metal powder **120B** is an alloy made of iron, silicon, and chromium and the average particle size of the magnetic metal powder **120B** is equal to or less than about 5.0 μm . In the embodiment, the first layer **L11** is

the lowermost layer in the up-down direction. That is, in the up-down direction, the side on which an outer electrode **230** is provided is referred to as an upper side, and the opposite side thereof is referred to as a lower side. The outer electrode **230** will be described later.

(102) A second layer **L12** having the substantially rectangular shape same as the first layer **L11** when viewed in the up-down direction is laminated on the upper side face of the first layer **L11** in the lamination direction as illustrated in FIG. **21**. In the embodiment, the face of the second layer **L12** contacting with the first layer **L11** is a main face **MF2** of the second layer **L12**. The second layer **L12** is constituted of a second magnetic layer **122**, a first inductor wiring line **130**, a first dummy wiring line **141**, a first connection wiring line **146**, and a first insulation portion **181**. The first inductor wiring line **130** is constituted of a first wiring line main body **131** having a substantially constant wiring line width, a first pad **132** connected to a first end of the first wiring line main body **131**, and a second pad **133** connected to a second end of the first wiring line main body **131**. The first inductor wiring line **130**, therefore, is laminated on the outer face of the first magnetic layer **121**.

(103) In the second layer **L12**, when viewed from the upper side in the up-down direction, the first wiring line main body **131** of the first inductor wiring line **130** spirally extends around the vicinity of the center of the face of the second layer **L12** having a substantially rectangular shape on the side opposite to the main face **MF2** as illustrated in FIG. **22**. Specifically, the first wiring line main body **131** of the first inductor wiring line **130** is spirally wound in a clockwise direction from the first end in the outer side portion in the radial direction toward the second end in the inner side portion in the radial direction.

(104) An angle in which the first inductor wiring line **130** is wound is about 540 degrees in the embodiment. With this, the number of turns of the wound first inductor wiring line **130** is about 1.5 turns in the embodiment. In addition, when the second layer **L12** is viewed from the upper side in the up-down direction, the side on which the first end of the first wiring line main body **131** is disposed is referred to as a first end side, and the side on which the second end of the first wiring line body **131** is disposed is referred to as a second end side in a long-side direction of the second layer **L12** having a substantially rectangular shape in the embodiment.

(105) The first pad **132** is connected to the first end on one side of the extending direction of the first wiring line main body **131**. The first pad **132** has a substantially rectangular shape when viewed from the upper side in the up-down direction. The first pad **132** constitutes a first end portion of the first inductor wiring line **130**. The first pad **132** is disposed in the vicinity of the corner of the second layer **L12** having a substantially rectangular shape when viewed in the up-down direction. The wiring line width of the first pad **132** is larger than the wiring line width of the first wiring line main body **131** connected to the first pad **132**.

(106) The second pad **133** is connected to the second end on the other side in the extending direction of the first wiring line main body **131**. The second pad **133** has a substantially circular shape when viewed from the upper side in the up-down direction. The second pad **133** constitutes a second end portion of the first inductor wiring line **130**. The diameter of the circle of the second pad **133** is larger than the width of the first wiring line main body **131** connected to the second pad **133**.

(107) The first dummy wiring line **141** is connected to the first pad **132**. The first dummy wiring line **141** extends from the portion of the first pad **132** on the side opposite to the first wiring line main body **131** toward the side face of the second layer **L12**, and is exposed to the outer face of the inductor component **110**.

(108) In the second layer **L12**, when viewed from the upper side in the up-down direction, the first connection wiring line **146** is disposed on the side opposite to the first pad **132** in a short-side direction of the second layer **L12** having a substantially rectangular shape and in the vicinity of the corner on the first end side in the long-side direction. The first connection wiring line **146** has the same shape as the first pad **132** and the first dummy wiring line **141**, and is line-symmetric with the

first pad **132** and the first dummy wiring line **141**. The symmetric axis is a straight line passing through the center of the second layer **L12** in the short-side direction and extending in the long-side direction of the second layer **L12**.

(109) The first inductor wiring line **130** has a laminated structure including a first wiring line layer **130B** and a second wiring line layer **130C** in order from the side of the first magnetic layer **121** constituting the first layer **L11** as illustrated in FIG. 23. The first wiring line layer **130B** of the first inductor wiring line **130** is in contact with the upper face of the first magnetic layer **121**, and constitutes most of the face of the first inductor wiring line **130** constituting the main face **MF2** of the second layer **L12**.

(110) The material of the first wiring line layer **130B** has a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt %. A thickness **TB2** of the first wiring line layer **130B** is equal to or less than about one-tenth of the wiring line width of the inductor wiring line **30**. The thickness **TB2** of the first wiring line layer **130B** is about 2.0 μm in this embodiment. The thickness **TB2** of the first wiring line layer **130B** is determined as follows. The thickness from the upper end of the first magnetic layer **121** to the upper end of the first wiring line layer **130B** in a cross section along the lamination direction is measured at three points in one observation field under a microscope of 1500 times magnification. The thickness **TB2** of the first wiring line layer **130B** is determined as the average value of the three measured values. The thickness **TB2** of the first wiring line layer **130B** is substantially constant in the embodiment. Note that the wiring line width of the first inductor wiring line **130** is determined as the average value of three points of the width dimension of the first inductor wiring line **130** in the vicinity of the center in the extending direction.

(111) The second wiring line layer **130C** is laminated on the upper side surface of the first wiring line layer **130B** on the side opposite to the first magnetic layer **121**. Further, the second wiring line layer **130C** covers an area slightly wider than the first wiring line layer **130B** from the upper side in the lamination direction. That is, the side face of the surface of the first wiring line layer **130B** facing the direction orthogonal to the lamination direction is covered by the second wiring line layer **130C**. Part of the outer face of the second wiring line layer **130C** constitutes part of the face of the first inductor wiring line **130** constituting the main face **MF2** of the second layer **L12**.

(112) A thickness **TC2** of the second wiring line layer **130C** is equal to or about five times larger than the thickness **TB2** of the first wiring line layer **130B**. The thickness **TC2** of the second wiring line layer **130C** is about 45 μm in the embodiment. Thus, the thickness of the first inductor wiring line **130** constituted of the first wiring line layer **130B** and the second wiring line layer **130C** is about 47 μm . The thickness **TC** of the second wiring line layer **130C** is determined as follows. The thickness from the upper end of the first wiring line layer **130B** to the upper end of the second wiring line layer **130C** in a cross section including the lamination direction is measured at three points in one observation field under a microscope of 1500 times magnification. The thickness **TC** of the second wiring line layer **130C** is determined as the average value of the three measured values. The material of the second wiring line layer **130C** has a copper ratio equal to or larger than about 99 wt %, and the nickel ratio is equal to or less than the detection limit.

(113) An anchor portion **134** extends from the main face **MF2** of the first inductor wiring line **130** as illustrated in FIG. 24. The anchor portion **134** extends from both of the first wiring line layer **130B** and the second wiring line layer **130C** constituting the main face **MF2** of the first inductor wiring line **130** in the embodiment. The anchor portion **134** covers the surfaces of the magnetic metal powders **120B** in contact with the main face **MF2** among the large number of magnetic metal powders **120B** in the first magnetic layer **121**. The anchor portion **134** extends from the main face **MF2** so as to enter a gap between the base material **120A** and the magnetic metal powder **120B** in the first magnetic layer **121**. Further, the magnetic metal powder **120B** covered by the anchor portion **134** includes a cross section in which equal to or larger than about one-third of the surface is covered by the anchor portion **134** when the magnetic metal powder **120B** is viewed in a cross

section.

(114) In the second layer L12, the side face of the first inductor wiring line 130, the side face of the first dummy wiring line 141, and the side face of the first connection wiring line 146 are covered by the first insulation portion 181 as illustrated in FIG. 22. That is, the first inductor wiring line 130, the first dummy wiring line 141, and the first connection wiring line 146 are surrounded by the first insulation portion 181. The first insulation portion 181 is insulation resin with an insulation property, and the insulation performance thereof is higher than that of the first inductor wiring line 130. Further, the first insulation portion 181 contains an inorganic filler. The portion other than the first inductor wiring line 130, the first dummy wiring line 141, the first connection wiring line 146, and the first insulation portion 181 is the second magnetic layer 122. Therefore, the second magnetic layer 122 is disposed in the central portion of the second layer L12, the both end portions of the second layer L12 in the short-side direction, and the first end side portion of the second layer L12 in the long-side direction. The material of the second magnetic layer 122 is the same as that of the first magnetic layer 121. As described above, since the first layer L11 is constituted of only the first magnetic layer 121, the lower face of the first inductor wiring line 130 is in contact with the first magnetic layer 121 without the first insulation portion 181 interposed therebetween.

(115) A third layer L13 having the substantially rectangular shape same as the second layer L12 when viewed in the up-down direction is laminated on the upper face of the second layer L12. The third layer L13 is constituted of a second insulation portion 182, a first via 191, a second via 192, a third via 193, and a third magnetic layer 123.

(116) The first via 191 is disposed on the upper side of the first pad 132 of the second layer L12, and is connected to the first pad 132. The second via 192 is disposed on the upper side of the first connection wiring line 146 of the second layer L12, and is connected to the first connection wiring line 146. The third via 193 is disposed on the upper side of the second pad 133 of the second layer L12, and is connected to the second pad 133. The first via 191, the second via 192, and the third via 193 have a substantially columnar shape, and the axial direction thereof coincides with the lamination direction. The length of the first via 191, the second via 192, and the third via 193 in the lamination direction is the same as the thickness of the third layer L13 in the lamination direction. Thus, the first via 191, the second via 192, and the third via 193 penetrate through the third magnetic layer 123 in the lamination direction.

(117) The second insulation portion 182 covers the first inductor wiring line 130, the first dummy wiring line 141, the first connection wiring line 146, and the first insulation portion 181 from the upper side. That is, the second insulation portion 182 covers all the face, of the upper face of the respective wiring lines disposed in the second layer L12, other than the portions where the first via 191, the second via 192, and the third via 193 are disposed. The second insulation portion 182 has a shape to cover an area slightly wider than the outer edges of the first inductor wiring line 130, the first dummy wiring line 141, and the first connection wiring line 146 when viewed from the upper side in the up-down direction. The second insulation portion 182 is insulation resin with an insulation property similar to the first insulation portion 181, and contains an inorganic filler. Note that a first insulation layer 180 is constituted of the first insulation portion 181 and the second insulation portion 182 in the embodiment. That is, of the surface of the first inductor wiring line 130, the portion of the upper side face in the lamination direction and of the face on the second magnetic layer 122 side excluding the first via 191 and the third via 193 is covered by the first insulation layer 180, and is in contact with the first insulation layer 180 in the embodiment.

(118) In the third layer L13, the portion excluding the first via 191, the second via 192, the third via 193, and the second insulation portion 182 is the third magnetic layer 123. Therefore, the third magnetic layer 123 is disposed in the central portion of the third layer L13, the both end portions of the third layer L13 in the short-side direction, and the first end side portion of the third layer L13 in the long-side direction. The third magnetic layer 123 is made of the same magnetic material as the first magnetic layer 121 described above.

(119) A fourth layer L14 having the substantially rectangular shape same as the third layer L13 when viewed in the up-down direction is laminated on the upper face of the third layer L13. The fourth layer L14 is constituted of a second inductor wiring line 135, a second dummy wiring line 142, a second connection wiring line 147, a third insulation portion 183, and a fourth magnetic layer 124. The second inductor wiring line 135 is constituted of a second wiring line main body 136 having a substantially constant wiring line width, a third pad 137 connected to a first end of the second wiring line main body 136, and a fourth pad 138 connected to a second end of the second wiring line main body 136. That is, the second inductor wiring line 135 is disposed on the upper side main face of the third magnetic layer 123 on the side opposite to the first inductor wiring line 130, and is laminated on the first inductor wiring line 130 with an interval of the third layer L13 in the lamination direction. Further, the third pad 137 serves as a first end portion of the second inductor wiring line 135, and the fourth pad 138 serves as a second end portion of the second inductor wiring line 135 in the embodiment.

(120) In the fourth layer L14, when viewed from the upper side in the up-down direction, the second wiring line main body 136 of the second inductor wiring line 135 spirally extends around the vicinity of the center of the face of the fourth layer L14 having a substantially rectangular shape on the side opposite to a main face MF3. Specifically, the second wiring line main body 136 of the second inductor wiring line 135 is spirally wound in the counterclockwise direction from the first end in the outer side portion in the radial direction toward the second end in the inner side portion in the radial direction. That is, the winding direction of the second inductor wiring line 135 is opposite to the winding direction of the first inductor wiring line 130.

(121) The angle in which the second inductor wiring line 135 is wound is about 540 degrees in the embodiment. With this, the number of turns of the wound second inductor wiring line 135 is about 1.5 turns in the embodiment.

(122) The third pad 137 is connected to the first end on one side of the extending direction of the second wiring line main body 136. The third pad 137 has a substantially rectangular shape when viewed in the up-down direction. The third pad 137 constitutes a first end portion of the second inductor wiring line 135. The third pad 137 is disposed in the vicinity of the corner of the fourth layer L14 having a substantially rectangular shape when viewed in the up-down direction. The third pad 137 is wider than the second wiring line main body 136 connected to the third pad 137 in the wiring line width.

(123) The fourth pad 138 is connected to a second end on the other side in the extending direction of the second wiring line main body 136. The fourth pad 138 has a substantially circular shape when viewed in the up-down direction. The fourth pad 138 is positioned on the upper side of the second pad 133 in the second layer L12, and is connected to the second pad 133 through the third via 193. The fourth pad 138 is wider than the second wiring line main body 136 connected to the fourth pad 138 in the wiring line width. The fourth pad 138 constitutes a second end portion of the second inductor wiring line 135.

(124) The second dummy wiring line 142 is connected to the third pad 137. The second dummy wiring line 142 extends from the portion of the third pad 137 on the side opposite to the second wiring line main body 136 toward the side face of the fourth layer L14, and is exposed to the outer face of the second inductor wiring line 135.

(125) In the fourth layer L14, when viewed from the upper side in the up-down direction, the second connection wiring line 147 is disposed on the side opposite to the third pad 137 in the short-side direction of the fourth layer L14 having a substantially rectangular shape and in the vicinity of the corner of the first end side in the long-side direction. The second connection wiring line 147 has the same shape as the third pad 137 and the second dummy wiring line 142, and is line-symmetric with the third pad 137 and the second dummy wiring line 142. The symmetric axis is a straight line passing through the center of the fourth layer L14 in the short-side direction and extending in the long-side direction of the fourth layer L14. The second inductor wiring line 135 and the second

connection wiring line **147** are illustrated by a broken line in FIG. **22**.

(126) The third via **193** is integrated with the second inductor wiring line **135** as illustrated in FIG. **23**. Although not illustrated in the drawings, the second via **192**, the second dummy wiring line **142**, and the second inductor wiring line **135** are integrated with one another. Further, the second connection wiring line **147** and the first via **191** are integrated with each other. The integrated object above described will be referred to as a second conductive layer **200** in the following description. The second conductive layer **200** has a laminated structure including a third wiring line layer **200A** and a fourth wiring line layer **200B**. The third wiring line layer **200A** constitutes part of the lower end side of the second conductive layer **200**. Therefore, the portion of the third wiring line layer **200A** positioned on the lower side of the first via **191** and the third via **193** is in contact with the first inductor wiring line **130**. In addition, the portion of the third wiring line layer **200A** positioned on the lower side of the second via **192** is in contact with the first connection wiring line **146**. Further, the portion of the third wiring line layer **200A** positioned on the lower side of other than the first via **191**, the second via **192**, and the third via **193** is in contact with the upper face of the second insulation portion **182**. The material of the third wiring line layer **200A** contains titanium and chromium.

(127) The fourth wiring line layer **200B** is laminated on the upper side surface of the third wiring line layer **200A** on the side opposite to the third magnetic layer **123**. The material of the fourth wiring line layer **200B** has a copper ratio equal to or larger than about 99 wt %. The upper end of the fourth wiring line layer **200B** is flush with the upper end of the fourth layer **L14**.

(128) In the fourth layer **L14**, the gap between the side faces of the second inductor wiring line **135** is covered by the third insulation portion **183** as illustrated in FIG. **21**. Therefore, the third insulation portion **183** is interposed at the portion where the distance between the wiring lines in the second inductor wiring line **135** is the shortest. The third insulation portion **183** is insulation resin with an insulation property, and the insulation performance thereof is higher than that of the second inductor wiring line **135**. In addition, the third insulation portion **183** does not contain an inorganic filler, unlike the first insulation layer **180**. The shape of the third insulation portion **183** is a curved shape as a whole.

(129) The portion other than the second inductor wiring line **135**, the second dummy wiring line **142**, the second connection wiring line **147**, and the third insulation portion **183** is the fourth magnetic layer **124**. Therefore, the fourth magnetic layer **124** is disposed in the central portion of the fourth layer **L14**, the both end portions of the fourth layer **L14** in the short-side direction, and the first end side portion of the fourth layer **L14** in the long-side direction. The material of the fourth magnetic layer **124** is the same as that of the first magnetic layer **121**.

(130) A fifth layer **L15** having the substantially rectangular shape same as the fourth layer **L14** when viewed in the up-down direction is laminated on the upper face of the surface of the fourth layer **L14**. The fifth layer **L15** is constituted of a fifth magnetic layer **125**, a fourth insulation portion **184**, a first columnar wiring line **194**, a second columnar wiring line **195**, and a third columnar wiring line **196**. The fifth magnetic layer **125** is disposed on the upper side of the second inductor wiring line **135** and the fourth magnetic layer **124** opposite to the first inductor wiring line **130**. The first columnar wiring line **194**, the second columnar wiring line **195**, and the third columnar wiring line **196** penetrate through the fifth layer **L15** in the lamination direction, that is, penetrate through the fifth layer **L15** from the face on the fourth magnetic layer **124** side toward the face on the side opposite to the fourth magnetic layer **124** side face. These columnar wiring lines function as vertical wiring lines in the embodiment.

(131) The fourth insulation portion **184** is directly connected to the upper side face of the third insulation portion **183** without any other layers interposed therebetween. The fourth insulation portion **184** covers a wider area than the third insulation portion **183** in the width direction orthogonal to the extending direction of the third insulation portion **183**. The thickness of the fourth insulation portion **184** is identical with the thickness of the fifth layer **L15**. The fourth insulation

portion **184** is insulation resin with an insulation property similar to the third insulation portion **183**, and the insulation performance thereof is higher than that of the second inductor wiring line **135**. Note that the second insulation layer **185** is constituted of the third insulation portion **183** and the fourth insulation portion **184** in the embodiment. The second insulation layer **185** is a non-magnetic material in contact with part of the surface of the second inductor wiring line **135**. Note that the face of the surface of the second inductor wiring line **135** on the first inductor wiring line **130** side is in contact with the first insulation layer **180**, and the face of the surface of the second inductor wiring line **135** on the fourth magnetic layer **124** side is in contact with the second insulation layer **185**.

(132) In the fifth layer **L15**, the portion excluding the first columnar wiring line **194**, the second columnar wiring line **195**, the third columnar wiring line **196**, and the fourth insulation portion **184** is the fifth magnetic layer **125**. The material of the fifth magnetic layer **125** is the same as that of the first magnetic layer **121** described above and is a magnetic material.

(133) A sixth layer **L16** having the substantially rectangular shape same as the fifth layer **L15** when viewed in the up-down direction is laminated on the upper face of the fifth layer **L15**. The sixth layer **L16** is constituted of a sixth magnetic layer **126**, a fourth columnar wiring line **197**, a fifth columnar wiring line **198**, and a sixth columnar wiring line **199**.

(134) The fourth columnar wiring line **197** is disposed on the upper side of the second connection wiring line **147** in the fourth layer **L14**, and is connected to the second connection wiring line **147** through the second columnar wiring line **195**. The sixth columnar wiring line **199** is disposed on the upper side of the third pad **137** of the fourth layer **L14**, and is connected to the third pad **137** through the first columnar wiring line **194**. The fourth columnar wiring line **197** and the sixth columnar wiring line **199** have a substantially prismatic shape, and the axial direction thereof coincides with the lamination direction. The length of the fourth columnar wiring line **197** and the sixth columnar wiring line **199** in the lamination direction is identical with the thickness of the sixth layer **L16** in the lamination direction. Thus, the fourth columnar wiring line **197** and the sixth columnar wiring line **199** penetrate through the sixth layer **L16** in the lamination direction. That is, the first columnar wiring line **194** and the sixth columnar wiring line **199** constitute a first vertical wiring line in the embodiment. Further, the second columnar wiring line **195** and the fourth columnar wiring line **197** constitute a third vertical wiring line.

(135) In addition, the fifth columnar wiring line **198** is disposed on the upper side of the fourth pad **138** of the second inductor wiring line **135** in the fourth layer **L14**, and is connected to the fourth pad **138** through the third columnar wiring line **196**. That is, the third columnar wiring line **196** and the fifth columnar wiring line **198** constitute a second vertical wiring line in the embodiment. Note that the fourth columnar wiring line **197**, the fifth columnar wiring line **198**, and the sixth columnar wiring line **199** are illustrated by a dashed-and-double-dotted line in FIG. 22.

(136) In the sixth layer **L16**, the portion excluding the fourth columnar wiring line **197**, the fifth columnar wiring line **198**, and the sixth columnar wiring line **199** is the sixth magnetic layer **126** as illustrated in FIG. 21. Thus, the sixth magnetic layer **126** is laminated on the upper side of the second inductor wiring line **135**. The material of the sixth magnetic layer **126** is the same as that of the first magnetic layer **121** described above and is a magnetic material.

(137) The outer electrode **230** is laminated on the upper side face of the fifth columnar wiring line **198** as illustrated in FIG. 23. Further, the outer electrode **230** is connected to the upper side face of the fourth columnar wiring line **197** and the sixth columnar wiring line **199**. Note that the outer electrode **230** is not illustrated in FIG. 21.

(138) Next, a manufacturing method of the inductor component **110** in the second embodiment will be described.

(139) As illustrated in FIG. 25, the manufacturing method of the inductor component **110** includes a first magnetic layer processing step, a first covering step, a first wiring line layer processing step, a first resist layer removing step, a second covering step, a second wiring line layer processing step,

a second resist layer removing step, and a first insulation layer processing step, and the first inductor wiring line **130** is formed through the steps described above. Further, the manufacturing method of the inductor component **110** includes a third wiring line layer processing step, a third covering step, a fourth wiring line layer processing step, a fourth covering step, a vertical wiring line processing step, a fourth resist layer removing step, a third resist layer removing step, a second insulation layer processing step, a second magnetic layer processing step, a base substrate removing step, an outer electrode processing step, and a dividing step, and the second inductor wiring line **135** and the like are formed through the steps described above.

(140) In manufacturing the inductor component **110**, first, the first magnetic layer processing step is performed. A base substrate with a copper foil **210** is prepared as illustrated in FIG. **26**. A base substrate **211** of the base substrate with the copper foil **210** has a plate-shape. A copper foil **212** is laminated on the upper side face of the base substrate **211** in the lamination direction. Then, the first magnetic layer **121** composed of the base material **120A** and the magnetic metal powder **120B** is formed on the upper side face of the copper foil **212** of the base substrate with copper foil **210** as illustrated in FIG. **27**. In forming the first magnetic layer **121**, an insulation resin containing the magnetic metal powder **120B** is applied, and the insulation resin is solidified by press working to obtain the base material **120A**. Then, the upper portions of the base material **120A** and the magnetic metal powder **120B** are ground such that the thickness in the up-down direction of the first magnetic layer **121** becomes a desired thickness. When the grinding is performed, it is preferable to form a slight gap at the interface between the base material **120A** and the magnetic metal powder **120B** by controlling process parameters during grinding.

(141) The first covering step is performed next to the second magnetic layer processing step. In the first covering step, on the upper side face of the first magnetic layer **121**, the first resist layer **221** covering the portion on which the first wiring line layer **130B** is not formed is patterned as illustrated in FIG. **28**. Specifically, first, the photosensitive dry film resist is applied to the entire upper side face of the first magnetic layer **121**. Next, on the upper side face of the first magnetic layer **121**, the portion on which the first wiring line layer **130B** is not formed is exposed to light. As a result, on the applied dry film resist, the portion exposed to light is cured. Thereafter, the uncured portion of the applied dry film resist is removed by a chemical solution. Thus, on the applied dry film resist, the cured portion is formed as the first resist layer **221**. On the other hand, the first magnetic layer **121** is exposed in the portion where the applied dry film resist is removed by a chemical solution and is not covered by the first resist layer **221**.

(142) The first wiring line layer processing step is performed next to the first covering step. In the first wiring line layer processing step, the first wiring line layer **130B** is formed on the upper side face of the first magnetic layer **121** as illustrated in FIG. **29**. Specifically, the electroless copper plating by performing immersion in the electroless copper plating solution forms the first wiring line layer **130B** having a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt % on the upper side face of the first magnetic layer **121** exposed from the first resist layer **221**. The electroless copper plating solution is an alkaline solution, and contains copper salts such as copper chloride, copper sulfate and the like. On the other hand, the material of the magnetic metal powder **120B** is iron, and the ionization tendency is larger than that of copper being the material of the first wiring line layer **130B**. Therefore, in the first wiring line layer processing step, iron on the surface of the magnetic metal powder **120B** dissolves, and instead, copper is deposited on the surface of the magnetic metal powder **120B**.

(143) Here, since the electroless copper plating solution enters a slight gap between the base material **120A** and the magnetic metal powder **120B**, the substitution of iron with copper occurs not only on the exposed face side of the magnetic metal powder **120B** but also on the surface of the magnetic metal powder **120B** on the inner side of the base material **120A**. Then, copper deposited on the surface of the magnetic metal powder **120B** on the inner side of the base material **120A** functions as the anchor portion **134**. Thus, the anchor portion **134** extending from the lower side

face of the first wiring line layer **130B** is formed with electroless copper plating.

(144) The first resist layer removing step for removing the first resist layer **221** is performed next to the first wiring line processing step. In the first resist layer removing step, the first resist layer **221** is peeled off to remove from the first magnetic layer **121** as illustrated in FIG. **30**.

(145) The second covering step is performed next to the first resist layer removing step. In the second covering step, on the upper side face of the first magnetic layer **121**, the second resist layer **222** covering the portion on which the second wiring line layer **130C** is not formed is patterned as illustrated in FIG. **31**. The second resist layer **222** is patterned such that an area slightly wider than the first wiring line layer **130B** is exposed in the embodiment. Note that the aspect of the photolithography in the second covering step is the same as that in the first covering step and a detailed description thereof will be omitted.

(146) The second wiring line layer processing step is performed next to the second covering step. In the second wiring line layer processing step, the second wiring line layer **130C** is formed in the portion not covered by the second resist layer **222**. Specifically, electrolytic copper plating is performed, and the second wiring line layer **130C** having a copper ratio equal to or larger than about 99 wt % is formed on the surface not covered by the second resist layer **222**. At this time, the end portion of the second wiring line layer **130C** is in direct close contact with the first magnetic layer **121** not covered by the first wiring line layer **130B** as illustrated in FIG. **24**. Therefore, the plating solution for the electrolytic copper plating enters the gap between the base material **120A** and the magnetic metal powder **120B** in the first magnetic layer **121** being in contact with the lower side face of the second wiring line layer **130C**. Copper precipitating from the plating solution and caught in the gap functions as the anchor portion **134**. The first wiring line processing step and the second wiring line processing step described above are the inductor wiring line processing steps in the embodiment.

(147) The second resist layer removing step is performed next to the second wiring line layer processing step. In the second resist layer removing step, the second resist layer **222** is peeled off to remove from the first magnetic layer **121** as illustrated in FIG. **32**.

(148) The first insulation layer processing step is performed next to the second resist layer removing step. The first inductor wiring line **130** is covered by an insulation material from the upper side in the lamination direction as illustrated in FIG. **33**. With this, the first insulation layer including the first insulation portion **181** and the second insulation portion **182** is formed on the entire upper side face of the first magnetic layer **121** and the first inductor wiring line **130**.

(149) The third wiring line processing step is performed next to the first insulation layer processing step. First, on the upper side face of the first inductor wiring line **130**, a hole penetrating through the second insulation portion **182** is formed by laser at the portion on which the third via **193** is formed as illustrated in FIG. **34**. With this, the upper side face of the first inductor wiring line **130** is exposed at the portion on which the third via **193** is formed. Next, the third wiring line layer **200A** functioning as a seed layer is formed by sputtering from the upper side in the lamination direction. The material of the third wiring line layer **200A** contains titanium and chromium.

(150) The third covering step is performed next to the third wiring line processing step. In the third covering step, on the surface of the third wiring line layer **200A**, the third resist layer **223** covering the portion on which the fourth wiring line layer **200B** is not formed is patterned. Note that the aspect of the photolithography in the third covering step is the same as that in the first covering step and a detailed description thereof will be omitted.

(151) The fourth wiring line layer processing step is performed next to the third covering step. In the fourth wiring line layer processing step, the electrolytic copper plating is performed, and on the surface of the third wiring line layer **200A**, the fourth wiring line layer **200B** having a copper ratio equal to or larger than about 99 wt % is formed on the portion not covered by the third resist layer **223**.

(152) The fourth covering step is performed next to the fourth wiring line processing step. In the

fourth covering step, the fourth resist layer **224** covering the portion in which the vertical wiring line is not formed is patterned as illustrated in FIG. **35**. That is, although not illustrated in the drawings, only the portions forming the first columnar wiring line **194**, the second columnar wiring line **195**, the third columnar wiring line **196**, the fourth columnar wiring line **197**, the fifth columnar wiring line **198**, and the sixth columnar wiring line **199** are exposed from the fourth resist layer **224**.

(153) The vertical wiring line processing step is performed next to the fourth covering step. In the vertical wiring line processing step, the electrolytic copper plating is performed, and on the surface of the second wiring line layer **130C**, each vertical wiring line of a copper ratio equal to or larger than about 99 wt % is formed on the portion not covered by the fourth resist layer **224**. That is, the third columnar wiring line **196** and the fifth columnar wiring line **198** are formed. Although not illustrated in the drawings, the first columnar wiring line **194**, the second columnar wiring line **195**, the fourth columnar wiring line **197**, and the sixth columnar wiring line **199** are also formed.

(154) The fourth resist layer removing step and the third resist layer removing step are performed at the same time next to the vertical wiring line processing step. Specifically, the third resist layer **223** and the fourth resist layer **224** are peeled off to remove from the first magnetic layer **121** as illustrated in FIG. **36**. Then, the third wiring line layer **200A** functioning as a seed layer exposed to the surface is removed by etching.

(155) The second insulation layer processing step is performed next to the third resist layer removing step. In the second insulation layer processing step, an insulation resin is applied to the upper side face as illustrated in FIG. **37**. Specifically, first, the insulation resin is applied from the upper side in the lamination direction to an extent in which the fourth wiring line layer **200B** is entirely covered. Next, the portion for forming the fourth insulation portion **184** is exposed to light. Thereafter, the uncured portion of the applied insulation resin is peeled and removed by a chemical solution. As a result, the portion of the applied insulation resin exposed to light is cured, and the third insulation portion **183** and the fourth insulation portion **184** are formed as illustrated in FIG. **38**. Thereafter, in the first insulation layer including the first insulation portion **181** and the second insulation portion **182**, the portion in which the first insulation portion **181** and the second insulation portion **182** are not formed, is removed by laser as illustrated in FIG. **39**.

(156) The second magnetic layer processing step is performed next to the second insulation layer processing step. In the second magnetic layer processing step, a magnetic material is filled to the upper side in the lamination direction relative to the upper end of the fifth columnar wiring line **198** as illustrated in FIG. **40**. Next, the grinding from the upper side in the lamination direction is performed until upper ends of the respective vertical wiring lines are exposed. With this, the second magnetic layer **122**, the third magnetic layer **123**, the fourth magnetic layer **124**, the fifth magnetic layer **125**, and the sixth magnetic layer **126** are formed.

(157) The base substrate removing step is performed next to the second magnetic layer processing step. In the base substrate removing step, the base substrate with the copper foil **210** is removed as illustrated in FIG. **41**. Specifically, the base substrate **211** is peeled off to remove from the first magnetic layer **121**. Next, the copper foil is removed by etching. Then, the first magnetic layer **121** is ground from the lower side in the lamination direction until the thickness from the lower end of the first magnetic layer **121** toward the upper end of the sixth magnetic layer **126** reaches a desired value.

(158) The outer electrode processing step is performed next to the base substrate removing step. Specifically, the outer electrode is formed on the upper side faces of the respective vertical wiring lines, that is, the upper side faces of the fourth columnar wiring line **197**, the fifth columnar wiring line **198**, and the sixth columnar wiring line **199** by electroless plating, electrolytic plating, printing, sputtering, or the like. The outer electrode has a single-layer or a laminated structure and includes any of copper, nickel, gold, and tin.

(159) The dividing step is performed next to the outer electrode processing step. Specifically, the

dividing is performed by cutting with a dicing machine at break lines DL as illustrated in FIG. 42. With this, the inductor component **110** may be obtained. In addition, at this time, the first dummy wiring line **141** and the second dummy wiring line **142** included in the break lines DL are exposed to the side face of the inductor component **110**.

(160) Next, effects of the second embodiment will be described. According to the second embodiment, in addition to the effects of (1) to (5) and (9) to (13) of the first embodiment, the following effects are achieved.

(161) (14) According to the inductor component **110** in the second embodiment, part of the second wiring line layer **130C** extends to the layer in which the first wiring line layer **130B** is disposed, and is interposed between the first wiring line layer **130B** and the second magnetic layer **122**. Therefore, the contact area between the first wiring line layer **130B** and the second wiring line layer **130C** is increased, and the adhesion between the first wiring line layer **130B** and the second wiring line layer **130C** is improved.

(162) (15) According to the inductor component **110** in the second embodiment, the entire fourth wiring line layer **200B** is laminated on the upper side face of the third wiring line layer **200A**, and is not interposed between the third wiring line layer **200A** and the fourth magnetic layer **124**. Therefore, different resist designs are allowed to manufacture the first inductor wiring line **130** and the second inductor wiring line **135**, and it improves the degree of freedom in design.

(163) (16) According to the inductor component **110** in the second embodiment, the first inductor wiring line **130** and the second inductor wiring line **135** are disposed in the lamination direction. That is, the inductor wiring line is not constituted as a single layer, but is constituted as a plurality of layers. Therefore, the inductance of the entire inductor component **110** may be improved.

(164) (17) According to the inductor component **110** in the second embodiment, the first insulation layer **180** contains insulation resin and an inorganic filler. Therefore, strength of the first insulation layer **180** may be improved.

(165) (18) According to the inductor component **110** in the second embodiment, the material of the first wiring line layer **130B** has a copper ratio equal to or less than about 99 wt % and a nickel ratio equal to or larger than about 0.1 wt %. Therefore, the first wiring line layer **130B** may be manufactured by electroless plating. Further, the third wiring line layer **200A** contains chromium or titanium. Therefore, the third wiring line layer **200A** may be manufactured by sputtering. As a result, the wiring line layers disposed in different layers may be manufactured by different manufacturing methods, which improves the degree of freedom in the manufacturing process.

(166) Each of the embodiments may be implemented with the following modifications. Each of the embodiments and the following modifications can be implemented in combination within a scope not technically contradicting each other.

(167) In each of the embodiments, the inductor wiring line is capable of imparting inductance to the inductor component by generating magnetic flux in the magnetic layer when an electric current flows.

(168) In each of the embodiments, the shape of the inductor wiring line is not limited to the example in each embodiment. For example, the inductor wiring line may have a curved shape less than about 1.0 turn or a linear shape of 0 turns. Further, part of a plurality of inductor wiring lines may have a shape different from that of other inductor wiring line. Furthermore, in each of the embodiments, the inductor wiring line may have a meander shape.

(169) In the first embodiment, a plurality of the inductor wiring lines **30** may be provided in the same layer. In this case, the provision of the plurality of inductor wiring lines **30** may suppress an excessive increase in the overall size in the lamination direction because the plurality of inductor wiring lines **30** is disposed in the same layer while an overall inductance is improved. Further, the inductor component **10** in which the plurality of inductor wiring lines **30** is provided in the same layer may be used by being divided into a plurality of inductor components.

(170) In each of the embodiments, the wiring line structure of the inductor wiring line is not limited

to the example in each embodiment. For example, in the inductor wiring line, the shapes of the first pad and the second pad may be changed, or the first pad and the second pad may be omitted.

(171) In the first embodiment, the catalyst layer **30A** and the second wiring line layer **30C** may be omitted in the inductor wiring line **30**, and the inductor wiring line **30** may be constituted of only the first wiring line layer **30B**. Even in this case, the lower side face of the first wiring line layer **30B** constitutes the main face MF of the inductor wiring line **30** and the anchor portion **34** needs to extend from the lower side face of the first wiring line layer **30B**.

(172) In each of the embodiments, the amount that the anchor portion covers is not limited to the example in each embodiment. For example, the anchor portion may not cover all of the surface of the magnetic metal powder in contact with, and may cover less than about one-third of the area thereof. In this case, the magnetic metal powder covered by the anchor portion may not include a cross section in which equal to or more than about one-third of the surface is covered by the anchor portion when the magnetic metal powder is viewed in a cross section. Further, the anchor portion may not cover the surfaces of all the magnetic metal powders in contact with the main face of the inductor wiring line. Furthermore, the anchor portion may be omitted.

(173) In each of the embodiments, the formation of the anchor portion and the control of the amount to be covered by the anchor portion are not limited to the example in the embodiment. For example, in the first embodiment, an alkaline chemical solution that dissolves the base material **20A** of the first magnetic layer **21** and does not dissolve the magnetic metal powder **20B** is used at the time of surface treatment such as removing the resin residue of the first magnetic layer **21**, and the interface state between the base material **20A** and the magnetic metal powder **20B** may be controlled with the duration of the processing time.

(174) In the first embodiment, a slight gap is formed between the base material **20A** and the magnetic metal powder **20B** when the grinding is performed and an electroless copper plating solution is infused into the gap in the inductor wiring line processing step. However, any other known method may be used as long as the anchor portion **34** may be formed. In particular, even in a case where there is no clear gap at the interface between the base material **20A** and the magnetic metal powder **20B**, the electroless copper plating solution enters along the interface between the base material **20A** and the magnetic metal powder **20B**, and the substitution of iron with copper described above is generated. Therefore, a gap between the base material **20A** and the magnetic metal powder **20B** need not be formed at the time of the grinding.

(175) In the first embodiment, the catalyst layer **30A** and the second wiring line layer **30C** may be omitted in the inductor wiring line **30**, and the inductor wiring line **30** may be constituted of only the first wiring line layer **30B**.

(176) In each of the embodiments, the material of the first wiring line layer is not limited to the example in each embodiment. For example, the material of the first wiring line layer may have a nickel ratio equal to or less than about 99 wt % and a phosphorus ratio equal to or larger than about 0.5 wt % and equal to or less than about 10 wt % (i.e., from about 0.5 wt % to about 10 wt %). In this case, containing phosphorus allows the control of the stress existing in nickel, and the residual stress in the inductor component may be mitigated. Further, containing nickel in the first wiring line layer may suppress electromigration.

(177) In each of the embodiments, the material of the second wiring line layer may be a metal other than copper. Note that the boundary face between the second wiring line layer and the first wiring line layer is not necessarily clear, and in some cases, a clear interface between the first wiring line layer and the second wiring line layer may not be observed.

(178) In each of the embodiments, the thickness of the second wiring line layer may be less than about five times the thickness of the first wiring line layer.

(179) In the first embodiment, the material of the catalyst layer **30A** is not limited to the example in the embodiment. The material of the catalyst layer **30A** needs to include at least one or more metals among palladium, platinum, silver, and gold.

(180) In the first embodiment, the thickness TA of the inductor wiring line **30** is not limited to the example in the embodiment. When the thickness TA of the inductor wiring line **30** is equal to or larger than about 40 μm , the DC resistance may be made relatively small. When the thickness TA of the inductor wiring line **30** is equal to or less than about 120 μm , the wiring line width with respect to the thickness TA may not excessively be increased.

(181) In the first embodiment, the thickness TB of the first wiring line layer **30B** is not limited to the example in the embodiment. When the thickness TB of the first wiring line layer **30B** is equal to or larger than about 0.3 μm and equal to or less than about 10 μm (i.e., from about 0.3 μm to about 10 μm), the first wiring line layer **30B** may easily be formed by electroless copper plating.

(182) In each of the embodiments, the material of the insulation layer is not limited to the example in each embodiment. For example, when the material of the insulation layer contains at least one resin among epoxy-based resin, phenol-based resin, acrylic-based resin, polyimide-based resin, and liquid crystal polymer-based resin, and an inorganic filler having an average particle size equal to or less than about 1 μm , it is suitable for ensuring the strength of the magnetic layer. Further, the material of the insulation layer is not limited to the above-description, and may be only a resin with an insulation property.

(183) In each of the embodiments, the area in which the insulation layer covers the inductor wiring line is not limited to the example in each embodiment. It is sufficient that at least the face of the surface of the inductor wiring line on the first magnetic layer side is not covered by the insulation layer and in contact with the first magnetic layer, and the face of the surface of the inductor wiring line on the second magnetic layer side is covered by the insulation layer. For example, in the first embodiment, all the face of the inductor wiring line **30** on the second magnetic layer **22** side may be covered by the insulation layer **80**. Further, all of the face of the inductor wiring line **30** on the third magnetic layer **23** side may be covered by the insulation layer **80**. Furthermore, the face of the inductor wiring line **30** on the third magnetic layer **23** side may not be covered by the insulation layer **80**. Still further, when part of the face of the inductor wiring line **30** on the second magnetic layer **22** side is covered by the insulation layer **80**, the insulation layer **80** may not be interposed at the portion where the distance between the wiring lines in the inductor wiring line **30** is minimum.

(184) In the second embodiment, the materials of the first inductor wiring line **130** and the second inductor wiring line **135** may be the same with each other.

(185) In the second embodiment, the fourth wiring line layer **200B** in the second inductor wiring line **135** may not extend to the layer in which the third wiring line layer **200A** is disposed, and may not be interposed between the third wiring line layer **200A** and the fifth magnetic layer **125**.

(186) In the manufacturing method in each of the embodiments, the dividing step may be omitted. In this case, for example, when the inductor component is manufactured in the size of one component from the first magnetic layer processing step, the dividing step may be omitted.

(187) In the embodiments, the boundaries of the magnetic layers in the respective layers may be integrated such that the interfaces cannot be confirmed, or may be separate bodies in which the interfaces can be confirmed.

(188) While preferred embodiments of the disclosure have been described above, it is to be understood that variations and modifications will be apparent to those skilled in the art without departing from the scope and spirit of the disclosure. The scope of the disclosure, therefore, is to be determined solely by the following claims.

Claims

1. An inductor component, comprising: a first magnetic layer; an inductor wiring line laminated on a main face of the first magnetic layer; a second magnetic layer disposed in a layer same as the inductor wiring line; a third magnetic layer disposed on a main face of the inductor wiring line and the second magnetic layer on a side opposite to the first magnetic layer; and an insulation layer

- being a non-magnetic material in contact with part of a surface of the inductor wiring line, wherein, an entire face of the surface of the inductor wiring line on the main face of the first magnetic layer is in physical contact with the first magnetic layer, and a face of the surface of the inductor wiring line on a side of the second magnetic layer is in contact with the insulation layer.
2. The inductor component according to claim 1, wherein a face of the surface of the inductor wiring line on a side of the third magnetic layer is in contact with the insulation layer.
 3. The inductor component according to claim 1, wherein part of the surface of the inductor wiring line is exposed from the insulation layer and is in contact with the second magnetic layer.
 4. The inductor component according to claim 1, wherein the inductor wiring line is wound more than 1.0 turns in the layer in which the inductor wiring line is disposed.
 5. The inductor component according to claim 4, wherein the insulation layer is interposed in a portion where a distance between wiring lines in the inductor wiring line is minimum in the layer in which the inductor wiring line is disposed.
 6. The inductor component according to claim 1, further comprising: a vertical wiring line being connected to the inductor wiring line and penetrating through the third magnetic layer from a main face on a side of the inductor wiring line toward a main face on a side opposite to a side of the inductor wiring line, wherein a face of a surface of the vertical wiring line on a side of the third magnetic layer is in contact with the insulation layer.
 7. The inductor component according to claim 1, wherein the insulation layer includes at least one resin among epoxy-based resin, phenol-based resin, acrylic-based resin, polyimide-based resin, or liquid crystal polymer-based resin.
 8. The inductor component according to claim 1, wherein the insulation layer contains an inorganic filler having an average particle size equal to or less than 1.0 μm .
 9. The inductor component according to claim 1, wherein the inductor wiring line has a laminated structure including a first wiring line layer on the main face of the first magnetic layer and a second wiring line layer laminated on a surface of the first wiring line layer on a side of the third magnetic layer, and part of the second wiring line layer extends to an inside of a layer in which the first wiring line layer is disposed, and is interposed between the first wiring line layer and the second magnetic layer.
 10. The inductor component according to claim 9, wherein the inductor wiring line includes a catalyst layer containing at least one or more metals among palladium, platinum, silver, or gold, and the catalyst layer is disposed on the first wiring line layer on the main face of the first magnetic layer.
 11. The inductor component according to claim 9, wherein a material of the first wiring line layer has a copper ratio equal to or less than 99 wt %, and a nickel ratio equal to or larger than 0.1 wt %.
 12. The inductor component according to claim 9, wherein a material of the first wiring line layer has a nickel ratio equal to or less than 99 wt %, and a phosphorus ratio from 0.5 wt % to 10 wt %.
 13. The inductor component according to claim 9, wherein the inductor wiring line has a thickness from 40 μm to 120 μm , and the first wiring line layer has a thickness from 0.3 μm to 10 μm .
 14. The inductor component according to claim 1, further comprising: a vertical wiring line being connected to the inductor wiring line and penetrating through the third magnetic layer from a main face on a side of the inductor wiring line toward a main face on a side opposite to a side of the inductor wiring line, wherein a covering layer formed of an insulation material is laminated on the main face of the third magnetic layer on a side opposite to the inductor wiring line, a face of the vertical wiring line on a side opposite to the inductor wiring line is not covered by the covering layer, and an outer electrode is connected to a portion in the vertical wiring line being exposed from the covering layer.
 15. The inductor component according to claim 1, wherein when the inductor wiring line is denoted as a first inductor wiring line and the insulation layer is denoted as a first insulation layer, the inductor component further comprises: a second inductor wiring line disposed on a main face of the

third magnetic layer on a side opposite to the first inductor wiring line; a fourth magnetic layer disposed in a layer same as the second inductor wiring line; a fifth magnetic layer disposed on the second inductor wiring line and the fourth magnetic layer on a side opposite to the first inductor wiring line; and a second insulation layer being a non-magnetic material in contact with part of a surface of the second inductor wiring line, wherein a face of the first inductor wiring line on a side of the third magnetic layer is in contact with the first insulation layer, a face of a surface of the second inductor wiring line on a side of the first inductor wiring line is in contact with the first insulation layer, and a face of the surface of the second inductor wiring line on a side of the fourth magnetic layer is in contact with the second insulation layer.

16. The inductor component according to claim 15, wherein the first insulation layer contains an inorganic filler, and the second insulation layer does not contain the inorganic filler.

17. The inductor component according to claim 16, wherein the first inductor wiring line has a laminated structure including a first wiring line layer and a second wiring line layer laminated on a surface of the first wiring line layer on a side opposite to the first magnetic layer, the second inductor wiring line has a laminated structure including a third wiring line layer and a fourth wiring line layer laminated on a surface of the third wiring line layer on a side opposite to the third magnetic layer, a material of the first wiring line layer has a copper ratio equal to or less than 99 wt %, and a nickel ratio equal to or larger than 0.1 wt %, and a material of the third wiring line layer includes at least one of chromium or titanium.

18. The inductor component according to claim 16, wherein the first inductor wiring line has a laminated structure including a first wiring line layer and a second wiring line layer laminated on a surface of the first wiring line layer on a side opposite to the first magnetic layer, the second inductor wiring line has a laminated structure including a third wiring line layer and a fourth wiring line layer laminated on a surface of the third wiring line layer on a side opposite to the third magnetic layer, part of the second wiring line layer extends to an inside of a layer in which the first wiring line layer is disposed, and is interposed between the first wiring line layer and the second magnetic layer, and the fourth wiring line layer is entirely laminated on a surface of the third wiring line layer on a side opposite to the third magnetic layer.

19. The inductor component according to claim 1, wherein the entire face of the surface of the inductor wiring line on the main face of the first magnetic layer is in direct contact with the first magnetic layer.

20. The inductor component according to claim 1, wherein a cross-section of the insulation layer formed in a shape of the letter T.

21. The inductor component according to claim 1, wherein an insulation layer being a non-magnetic material in contact with part of a surface of the inductor wiring line, the insulation layer extending for less than one turn between adjacent turns of the inductor wiring line.

22. An inductor component, comprising: a first magnetic layer; an inductor wiring line laminated on a main face of the first magnetic layer; a second magnetic layer disposed in a layer same as the inductor wiring line; a third magnetic layer disposed on a main face of the inductor wiring line and the second magnetic layer on a side opposite to the first magnetic layer; and an insulation layer being a non-magnetic material in contact with part of a surface of the inductor wiring line, wherein, an entire face of the surface of the inductor wiring line on the main face of the first magnetic layer is in contact with the first magnetic layer, a face of the surface of the inductor wiring line on a side of the second magnetic layer is in contact with the insulation layer, and a portion of the insulation layer is positioned directly between the inductor wiring line and the third magnetic layer.
